Test Result Summary

POSITIVE Clinically relevant actionable variant was detected in KRAS gene

TABLE 1: ALTERATIONS WITH STRONG CLINICAL PROGNOSTIC AND THERAPEUTIC SIGNIFICANCE IN PATIENT TUMOR TYPE

*Refer to attachment for Clinical Trials

TABLE 2: NON DRUGGABLE / DRUGGABLE ALTERATIONS WITH POTENTIAL CLINICAL AND THERAPEUTIC SIGNIFICANCE IN OTHER TUMOR TYPES

TABLE 3: VARIANTS OF UNKNOWN SIGNIFICANCE

*Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines. #Correlation of the genetic findings with the clinical condition of the patient is required to arrive at an accurate diagnosis, prognosis or therapeutic decisions.

Variant Classification as per AMP-ASCO-CAP Guidelines

Variant A change in a gene. This could be disease causing (pathogenic) or not disease causing (benign).

Tier I: Variants of Strong Clinical Significance

Level A , biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors; Level B , biomarkers that predict response or resistance to a therapy based on well- powered studies with consensus from experts in the field or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.

Tier II: Variants of Potential Clinical Significance

Level C , biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies. Level D , biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.

General Content / Notes

Oncq Sol Focus 52 Gene Panel is a multibiomarker next-generation sequencing (NGS) assay that enables detection of variants in 52 genes relevant to solid tumors. This assay allows concurrent analysis of DNA and RNA to simultaneously detect multiple types of variants, including hotspots, single-nucleotide variants (SNVs), indels, copy number variants (CNVs), and gene fusions , in a single workflow. Some of the alterations detected may provide prognostic/ therapeutic information that may help the Clinician in deciding upon the best line of Targeted/ Chemotherapy options for the patient.

Methodology

FFPE DNA was extracted from tissue using Qiagen Kit and FFPE RNA was extracted from tissue Promega Kit. The DNA and RNA were quantified on Qubit fluorometer. Amplicon based barcoded library was prepared using the extracted DNA and RNA. Libraries were pooled and loaded onto ION Torrent S5 NGS platform to perform massive parallel sequencing. The average Variant Allele Frequency (VAF) for this Test is 5% at 500X coverage. However, it may vary depending on the particular mutation type. Alterations outside of the tested areas of these genes or in other genes will not be detected. This Panel analyzed the following types of mutations- nucleotide substitutions, small insertions, small deletions, indels, CNVs & gene fusions. If the mutation(s) fall in the region of no/ low coverage zone, then they may be missed. Tumor sample must be fixed under appropriate conditions (10% Neutral buffered formalin for 12-72hrs) to ensure preservation of amplifiable quality DNA. PCR is a highly sensitive technique; reasons for apparently contradictory results may be due to improper quality control during sample collection and fixation, high degree of necrosis, mucin content and/or presence of PCR inhibitors. Cytosine deamination has been demonstrated to contribute to background noise for DNA sequencing of ancient and formalin fixed paraffin embedded (FFPE) treated DNA. Deamination can be attributed to multiple factors including biologic (intrinsic to the sample prior to isolation) or an artifact of the molecular biology. It has been observed that formalin induces C:G>T:A transitions during sequencing. These deamination related “apparent mutations” are higher in the FFPE samples than in peripheral blood indicating that the process of specimen

General Content / Notes

5000Exomes (Source Version:20161108), Canonical RefSeq Transcripts(Source Version:v95), ClinVar(Source Version:20190909), dbSNP(Source Version:153), DGV(Source Version:20160515), DrugBank(Source Version:20190723), ExAC(Source Version:1), Gene Ontology(Source Version:20190930), Named Variants(Source Version:1), Pfam(Source Version:32), PhyloP Scores(Source Version:20160919), RefSeq Gene Functional Canonical Transcripts Scores(Source Version:10), RefSeq GeneModel(Source Version:95), CitelineTrialTrove.

Limitation and Disclaimer

Variant with 250X coverage or more are considered for clinical correlation with the disease. This test is limited to listed genes analysis only, It should be noted that this test does not sequence all bases in a human genome, not all variants have been identified or interpreted, and this report is limited only to variants with evidence for causing or contributing to disease/clinical details. This assay is not meant to interrogate most promoter regions, deep intronic regions, or other regulatory elements. Data from this test is based on currently available scientific information. The classification of variants of unknown significance can change over time.This data can be re-assessed for the presence of any variants that may be newly linked to be associated with cancer. The physician can request reanalysis of the data. Certain genes may not be covered completely, and few mutations could be missed. This NGS test used does not allow definitive differentiation between germline and somatic variants. TREATMENT DECISIONS BASED ON THESE MUTATIONS MAY BE TAKEN IN CORRELATION WITH OTHER CLINICAL AND PATHOLOGICAL INFORMATION

All investigations have their limitation which is imposed by the limits of sensitivity and specificity of individual assay procedures as well as the quality of the specimen received at laboratory. Inclusion of variations is dependent upon our assessment of their significance. The quality of sequencing and coverage varies between regions. Many factors such as homopolymers, GC-rich regions etc. influence the quality of sequencing and coverage. This may result in an occasional error in sequence reads or lack of detection of a particular genetic lesion.

Analysis

Torrent Variant Caller plugin – A SNP and indel calling analysis module that is part of the Torrent Suite™ Software and is accessible through the Torrent Browser. The plugin can either be initiated automatically after sequencing data have been generated and bases called or it can be initiated manually to process previously generated datasets. Ion Reporter™ Software – A cloud-based software service that provides both variant calling and annotations, designed with traceability features that are essential to researchers who perform sequencing assays. It comprises a suite of bioinformatics tools that streamlines and simplifies analysis, reporting, and archiving of semiconductor sequencing data. Ion Reporter™ Software fast-tracks variant analysis and integrates comprehensive public annotations to reduce the bioinformatics work needed to understand the impact of detected variants and to shorten the time to interpret test results. Oncomine™ Knowledgebase Reporter - The Oncomine™ Reporter software matches genetic variant information from a source file with relevant data from a curated set of published evidence from public sources. The software is used to prepare a report that presents a sample-specific view of data with each variant matched with relevant data, including labels, guidelines, and clinical trials. Variants of Unknown Significance (VUS) – These variants may not have been adequately characterized in the scientific literature at the time this report was issued and/or the genomic context of these alterations makes their significance unclear. We choose to include them here in the event that they become clinically meaningful in the future.

General Content / Notes

Note: The performance of this Test has been evaluated at Oncquest Laboratories Ltd.

*** End Of Report ***

Gene Finding Gene Finding

BRAF None detected NTRK2 None detected ERBB2 None detected NTRK3 None detected KRAS KRAS G12V PIK3CA None detected NRAS None detected RET None detected NTRK1 None detected

KRAS G12V

Allele Frequency: 16.72%

bevacizumab + chemotherapy avutometinib + defactinib 1 14

* Public data sources included in relevant therapies: NCCN, ESMO, FDA 1

⍰ Alerts informed by public data sources: ⍰ Contraindicated, ⍰ Resistance, ⍰ Breakthrough, ⍰ Fast Track

KRAS proto-oncogene, GTPase

Background: The KRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes NRAS and HRAS 1 . RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival 2,3,4 . Germline mutations in KRAS lead to several genetic disorders known as RASopathies, including Noonan syndrome, which results in heart and congenital defects, growth inhibition, and facial dysmorphic features 5 . Somatic mutations in KRAS are commonly altered in several cancers including non-small cell lung cancer, pancreatic cancer, and multiple myeloma 5 .

Alterations and prevalence: The majority of KRAS mutations consist of point mutations occurring at G12, G13, and Q61 6,7,8 . Mutations at A59, K117, and A146 have also been observed but are less frequent 9,10 . Somatic mutations in KRAS are observed in 66% of pancreatic adenocarcinoma, 41% of colorectal adenocarcinoma, 30% of lung adenocarcinoma, 19% of uterine corpus endometrial carcinoma, 12% of uterine carcinosarcoma, 9% of stomach adenocarcinoma, 8% of testicular germ cell tumors, 6% of cholangiocarcinoma, 5% of cervical squamous cell carcinoma, acute myeloid leukemia, and diffuse large B-cell lymphoma, 4% of bladder urothelial carcinoma, and 2% of skin cutaneous melanoma and kidney renal papillary cell carcinoma 6,9 . KRAS is amplified in 9% of ovarian serous cystadenocarcinoma and testicular germ cell tumors, 8% of stomach adenocarcinoma , 7% of esophageal adenocarcinoma and uterine carcinosarcoma, 6% of lung adenocarcinoma, 4% of pancreatic adenocarcinoma and bladder urothelial carcinoma, 3% of lung squamous cell carcinoma, and 2% of sarcoma, mesothelioma, brain lower grade glioma, and uterine corpus endometrial carcinoma 6,9 . Alterations in KRAS are also observed in pediatric cancers 9 . Somatic mutations in KRAS are observed in 10% of B-lymphoblastic

General Content / Notes

Potential relevance: The FDA has approved the small molecule inhibitors, sotorasib 11 (2021) and adagrasib 12 (2022), for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic non-small cell lung cancer (NSCLC). Sotorasib and adagrasib are also useful in certain circumstances for KRAS G12C-mutated pancreatic adenocarcinoma 13 . The FDA has approved adagrasib 12 in combination with cetuximab (2024), and sotorasib 11 in combination with panitumumab 14 (2025) for KRAS G12C-mutated colorectal cancer (CRC) after prior fluoropyrimidine-, oxaliplatin-, and irinotecan-based chemotherapy. The EGFR antagonists, cetuximab 15 and panitumumab 14 , are contraindicated for treatment of colorectal cancer patients with KRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146) 10 . The FDA has approved the combination of kinase inhibitors, avutometinib and defactinib 16 (2025), for the treatment of adult patients with KRAS-mutated recurrent low-grade serous ovarian cancer (LGSOC) after prior systemic therapy. The FDA has granted breakthrough therapy designation (2022) to the KRAS G12C inhibitor, GDC-6036 17 , for KRAS G12C-mutated NSCLC. The KRAS-G12C/NRAS-G12C dual inhibitor, elironrasib 18 , and the KRAS G12C inhibitor, D3S-001 19 , were both granted breakthrough therapy designation (2025) for KRAS G12C-mutated locally advanced or metastatic NSCLC in adults previously treated with chemotherapy and immunotherapy, excluding KRAS G12C inhibitors. The KRAS- G12C inhibitor, olomorasib 20 , was granted breakthrough designation (2025) in combination with pembrolizumab 21 for unresectable advanced or metastatic NSCLC with a KRAS G12C mutation and PD-L1 expression ≥ 50%. The RAF/MEK clamp, avutometinib 22 was also granted fast track designation (2024) in combination with sotorasib for KRAS G12C-mutated metastatic NSCLC in patients who have received at least one prior systemic therapy and have not been previously treated with a KRAS G12C inhibitor. The KRAS G12C inhibitor, BBO-8520 23 , was granted fast track designation in 2025 for previously treated KRAS G12C-mutated patients with metastatic NSCLC. The RAS(ON) G12D-selective inhibitor, zoldonrasib 24 , was granted breakthrough designation (2026) for patients with KRAS G12D-mutated locally advanced or metastatic NSCLC who have been previously treated with anti-PD-1/PD-L1 therapy and platinum-based chemotherapy. The RAS inhibitor, daraxonrasib 25 , was granted breakthrough designation (2025) for previously treated metastatic pancreatic ductal adenocarcinoma (PDAC) with KRAS G12 mutations. The KRAS G12D (ON/OFF) inhibitor, GFH-375 26 , was also granted fast track designation (2025) for first-line and previously treated KRAS G12D-mutated locally advanced or metastatic PDAC. The pan-KRAS inhibitor, BBO-11818 27 , was also granted fast track designation (2026) for adult patients with advanced KRAS- mutant PDAC. The KRAS G12C inhibitor, D3S-001 28 , was granted fast track designation in 2024 for KRAS G12C-mutated patients with advanced unresectable or metastatic CRC. The PLK1 inhibitor, onvansertib 29 , was granted fast track designation (2020) in combination with bevacizumab and FOLFIRI for second-line treatment of patients with KRAS-mutated metastatic CRC. The immuno-oncolytic virus, pelareorep 30 , was also granted fast track designation (2026) in combination with bevacizumab and FOLFIRI for second-line treatment of patients with KRAS-mutated microsatellite-stable metastatic CRC. Additionally, KRAS mutations are associated with poor prognosis in NSCLC 31 . Mutations in the RAS pathway, including KRAS, are ancillary diagnostic markers in pediatric gene fusion-negative rhabdomyosarcoma (RMS) 32 .

NCT ID Title Phase

NCT06947811 A Multicenter, Open-Label, Single-Arm, Phase Ib/II Clinical Study of Almonertinib Combined With Palbociclib in the Treatment of Patients With KRAS Mutations-Positive Advanced Solid Tumors

I/II

NCT07353645 Phase I/II Clinical Study of KRAS Neoantigen Nanovaccine as Adjuvant Therapy for Colorectal Cancer/ Pancreatic Cancer With High Risk of Recurrence

I/II

NCT05173805 Phase I Clinical Study on the Safety, Tolerance, Pharmacokinetics and Efficacy of YL-15293 in Patients With Advanced Solid Tumor With KRAS Mutation

I/II

NCT07455617 A Phase I Clinical Study to Evaluate the Safety, Tolerability, Immunogenicity, Pharmacokinetics, Pharmacodynamics and Preliminary Anti-tumor Activity of ABO2102 in Participants With KRAS Mutant Solid Tumors

I

General Content / Notes

No NCT ID An open-label, multicenter Phase I clinical study to evaluate the safety, tolerability, pharmacokinetic characteristics, and immunogenicity of RGL-232 in patients with malignant solid tumors

I

NCT07004244 Safety and Efficacy of KRAS Vaccine in the Treatment of KRAS-mutant Malignancies I

NCT07435038 A Phase I/II Clinical Study Evaluating the Safety, Tolerability, Pharmacokinetics, and Efficacy of BPI-572270 Capsules in Patients with RAS-mutant Advanced Solid Tumors.

I/II

NCT06895031 A Multi-center, Open-label, Phase I/II Clinical Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Antitumor Activity of JYP0015 in Advanced Solid Tumors With RAS Mutation

I/II

NCT05369312 Phase I Study of BPI-442096 in Advanced Solid Tumor Patients I

NCT06962254 Treatment of Solid Tumors Harboring KRAS Mutation With Imatinib and Trametinib I

NCT05004441 Fruquintinib Combined With mFOLFOX6/FOLFIRI in First-line Treatment for Metastatic Colorectal Cancer:HCCSC C02 Trial

II

General Content / Notes

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Table 1 (Page 1)

Gene	CDS variant details	Amino acid change/ Exon No.	Overall depth/ Mutant Allele Percentage	Function of the gene in cancer	Variant Classification and its relevance in patient tumor type
KRAS	c.35G>T	p.Gly12Val/ Exon 2	1537x/ 16.72%	Gain-of-Function	Tier I:A

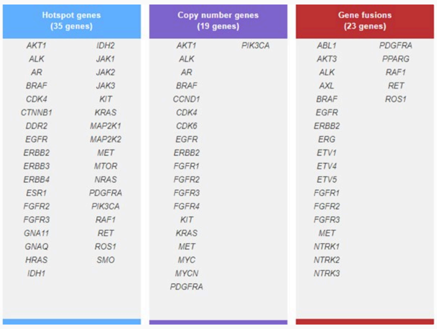
Table 2 (Page 1)

Gene	CDS variant details	Amino acid change/ Exon No.	Overall depth/ Mutant Allele Percentage	Function of the gene in cancer	Variant Classification and its relevance in other tumor type
None					

Table 3 (Page 1)

Gene	CDS variant details	Amino acid change/ Exon No.	Overall depth/ Mutant Allele Percentage	Function of the gene in patient tumor/ Other tumor type
None				

Extracted Body Images & Graphical Content



Hotspot genes (35 genes)		Copy number genes (19 genes)		Gene fusions (23 genes)	
AKT1	*IDH2*	*AKT1*	*PIK3CA*	*ABL1*	*PDGFRA*
ALK	*JAK1*	*ALK*		*AKT3*	*PPARG*
AR	*JAK2*	*AR*		*ALK*	*RAF1*
BRAF	*JAK3*	*BRAF*		*AXL*	*RET*
CDK4	*KIT*	*CCND1*		*BRAF*	*ROS1*
CTNNB1	*KRAS*	*CDK4*		*EGFR*	
DDR2	*MAP2K1*	*CDK6*		*ERBB2*	
EGFR	*MAP2K2*	*EGFR*		*ERG*	
ERBB2	*MET*	*ERBB2*		*ETV1*	
ERBB3	*MTOR*	*FGFR1*		*ETV4*	
ERBB4	*NRAS*	*FGFR2*		*ETV5*	
ESR1	*PDGFRA*	*FGFR3*		*FGFR1*	
FGFR2	*PIK3CA*	*FGFR4*		*FGFR2*	
FGFR3	*RAF1*	*KIT*		*FGFR3*	
GNA11	*RET*	*KRAS*		*MET*	
GNAQ	*ROS1*	*MET*		*NTRK1*	
HRAS	*SMO*	*MYC*		*NTRK2*	
IDH1		*MYCN*		*NTRK3*	
		PDGFRA			

Page 2 • Image (602×439px)

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